GREEN COMPUTE MRV

Strategic Green Compute Assessment

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1. Executive Summary: Optimization Pathways

Pathway	Target Region	Carbon Intensity	Total Emissions	Risk Profile
Baseline	West India (Maharashtra/Mumbai)	575 g/kWh	391000 gCO2	None
Domestic	North India (Delhi / NCR)	423 g/kWh	287640 gCO2	Low (Localized)
International	Nordics (Sweden - Hydro/Nuclear)	26 g/kWh	17680 gCO2	High (Data Export)

2. Strategic Advisory & Risk Analysis

Executive Briefing: Domestic vs. International Data Center Options

#1. Comparison of Domestic and International Options

In establishing data centers, organizations can choose between domestic and international operations. Each option carries distinct implications for emissions profiles, energy procurement, and regulatory compliance.

- **Domestic Options**:
- Generally aligned with national regulations, such as the India Carbon Credit Trading Scheme (CCTS), which encourages energy efficiency and provides a market for carbon credits.
- Offering lower emissions based on local grid factors, domestic operations exhibit significant energy savings (e.g., 103,360 grams versus 373,320 grams for international operations).
- Facilitate traceability and accountability in GHG emissions, supporting adherence to ISO 14064-1.

- **International Options**:
- Access to global talent pools and advanced technologies.
- However, operations in countries with laxer environmental regulations may pose challenges, particularly regarding compliance with organizations' sustainability standards.
- Potentially higher emissions due to less stringent energy sources and larger emissions footprints that

can conflict with corporate climate commitments.

#2. Data Sovereignty Risks for International Operations

The shift to international data center operations raises critical data sovereignty risks, including:
- **Compliance and Regulatory Challenges**: Differing international data protection laws may lead to complications in adhering to local regulations governing data storage and processing.
- **Data Breach Vulnerabilities**: Relying on international facilities may increase the risk of data breaches due to diverse cybersecurity protocols and legal frameworks, creating liabilities.
- **Volatile Energy Costs and Emissions Reporting**: Fluctuating international energy prices and emissions regulations can increase operational costs and complicate emissions reporting and compliance with frameworks like ISO 14064-1.

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#3. Recommended Hybrid Approach

To mitigate risks associated with both domestic and international options, a hybrid approach is recommended:
- **Leverage Domestic Capacities**: Position primary data processing in domestic centers that comply with local environmental regulations, maximizing energy efficiency and emissions reporting accuracy.
- **Supplement with International Facilities**: Utilize select international data centers for non-sensitive operations, research, or redundancy, ensuring alignment with organizational sustainability goals.
- **Optimize Energy Purchase Agreements (PPAs)**: Establish dedicated PPAs that source energy from verified renewable resources, regardless of location, further enhancing emissions profiles.
- **Standardize Reporting and Compliance**: Implement consistent metrics across all operations to ensure transparency and traceability of emissions in line with CCTS and ISO standards.

This hybrid model enables organizations to capitalize on the benefits of both operational frameworks while minimizing associated risks, fostering a sustainable data infrastructure aligned with corporate sustainability objectives.

CRITICAL COMPLIANCE NOTICE:
International routing requires adherence to Data Sovereignty laws (DPDP Act, GDPR). Ensure Standard Contractual Clauses (SCCs) are in place.